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Attribution of the record-breaking February 2022 compound wet and cold event over Southern China

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Abstract

In February 2022, Southern China experienced a severe compound wet and cold event (CWCE) that caused major socioeconomic impacts on transportation, power supply, and agriculture, ranking among China's top ten natural disasters of the year. To what extent anthropogenic climate change has already affected such CWCEs remained unclear. This study employs a circulation analogue-based risk attribution framework with observation to quantify contribution of large-scale atmospheric circulation and the thermodynamic effect of climate change to the event, and further used simulations from coupled model intercomparison project (CMIP6) models under different external forcing to assess the anthropogenic influence. The main findings are as follows: atmospheric circulation anomalies were the dominant driver of this CWCE. The contribution of circulation patterns similar to those in February 2022 was estimated at 73% (95% confidence intervals, CIs: 71%, 76%). Thermodynamic effect of climate change reduced the probability of occurrence of this CWCE by 32% (95% CIs: 27%, 36%). Further analysis using CMIP6 models and a risk-based attribution approach confirms that anthropogenic forcing suppresses the occurrence of CWCE, reducing its probability by 21% (95% CIs: 5%, 33%).

1. Introduction

In the context of ongoing global warming, the frequency, intensity, and duration of extreme weather and climate events have markedly increased, posing severe threats to global ecosystems, socio-economic systems, and human well-being (Seneviratne *et al* 2021, Yin *et al* 2024). Extreme events can exhibit compound characteristics, for example, multiple extreme events can occur simultaneously or sequentially in time or space. These are commonly referred to as compound extreme events (Zscheischler *et al* 2018, Ridder *et al* 2020). The complexity and synergistic effects of compound extreme events result in impacts far exceeding those of single events, necessitating urgent in-depth research to uncover their occurrence mechanisms, attribution analyses, and adaptation strategies (Zscheischler *et al* 2020, Ridder *et al* 2022). The intergovernmental panel on climate change (IPCC), in its 2012 special report on managing the risks of extreme events and disasters to advance climate change adaptation (SREX), systematically outlined the definition, classification, and significance of compound extreme events for climate adaptation (Seneviratne *et al* 2012). The SREX report provides a theoretical framework for studying compound events, underscoring their critical importance in risk assessment and disaster management.

Current research on compound extreme events primarily focuses on the combination of extreme high temperature with drought or heavy precipitation (Wolski *et al* 2014, Zhang and Villarini 2020, Chen *et al* 2021, Wu *et al* 2021, Pan *et al* 2023). In contrast, studies on compound events involving extreme cold, such as compound wet and cold events (CWCEs), are relatively scarce, with substantial knowledge gaps regarding their driving mechanisms and attribution (De Luca *et al* 2020, Li *et al* 2023). CWCEs are typically defined as the simultaneous or sequential occurrence of extreme cold events and extreme precipitation events within a short time window (e.g. days to weeks). These events may involve complex physical processes, such as the interaction between cold air intrusions and moisture transport, leading to phenomena like snowfall, freezing rain, or persistent low-temperature high-humidity conditions. CWCEs pose severe challenges to ecosystems (e.g. disrupted vegetation phenology and reduced biodiversity), socio-economic systems (e.g. supply chain disruptions and public health risks), and disaster management (Markantonis *et al* 2022, Messori and Faranda 2023). Although global warming is expected to reduce the frequency of cold extremes, emerging evidence suggests that CWCEs likelihood may increase in certain mid-latitude regions (Chen *et al* 2023, Nie *et al* 2025). This apparent paradox is linked to Arctic amplification, which weakens meridional temperature gradients and may contribute to a wavier polar jet stream, more persistent blocking patterns (e.g. Ural or Siberian blocking), and altered stratospheric–tropospheric coupling (Francis and Vavrus 2012, Cohen *et al* 2014, Yao *et al* 2017, Luo *et al* 2019). These dynamical changes can facilitate occasional but intense southward intrusions of Arctic air masses. When such intrusions coincide with enhanced moisture availability (Zheng *et al* 2022)—due to warmer oceans or shifted storm tracks—the risk of CWCE can be amplified in mid-latitude regions. This poses new and urgent challenges for mechanistic understanding, risk assessment, and adaptation planning.

In February 2022, southern China suffered a widespread low-temperature, rain, snow, and freezing disaster, ranked among China's top ten natural disasters of that years. In most regions of southern China, the monthly mean temperature was 2 °C–4 °C below the normal years (1991–2020), while the accumulated monthly precipitation was 50% to 200% above normal. Notably, from 17 to 29 February, the average depth of snow across much of southern China reached 14.2 mm, ranking as the second-highest on record for the same period. This event severely disrupted agricultural production, transportation networks, and power systems, impacting 6.09 million people across nine provinces, causing over 900 houses to collapse and nearly 6500 to be damaged, and resulting in direct economic losses of 7.89 billion yuan (https://politics.gmw.cn/2023-01/12/content_36297529.htm). Large-scale atmospheric circulation anomalies were the primary driver of this event (Ma *et al* 2022, Li *et al* 2023). The circulation anomalies were jointly triggered by cold sea surface temperature (SST) in the Tropical Pacific and warm SST in the North Atlantic, leads to a strengthening of the Siberian High, a deepening of the India-Burma trough, and the development of an anomalous anticyclone over the western North Pacific. Human activities also exert an influence on the occurrence of this compound event. Previous attribution studies have examined the individual low temperature event and heavy precipitation event in February 2022 separately. Human activities have made cold events similar to the February 2022 cold event 1.5 times less likely, with greenhouse gases (GHGs) being the main driver and aerosols exerting a counteracting influence (Chen *et al* 2023). Results regarding the attribution of the extreme precipitation event to human activities show discrepancies findings. Hu *et al* (2023) indicates that human activities reduced the likelihood of such an event by approximately 50%. While, Qiao *et al* (2023) highlight that due to model biases in simulating regional precipitation processes and the large influence of internal climate variability at regional scales, no detectable signal of human influence was found on this extreme precipitation event. To what extent anthropogenic climate change has already affected such CWCEs is thus critical for informing evidence-based adaptation strategies and policy development.

To address this gap, this study first uses observations and reanalysis data together with a circulation analogue method to separate and quantify the contributions of circulation anomalies and the thermodynamic effects of climate change to the event's intensity. We then employ model simulations with different external forcing to assess how much different anthropogenic forcing affects the probability of such events. This work helps fill the current shortage of attribution studies on CWCEs, improves our understanding of the changing characteristics of CWCEs under global warming, and provides a scientific basis for climate adaptation policies and risk management in southern China.

2. Data and method

2.1. Data

This study uses daily surface air temperature (SAT) and precipitation (Pre) observations spanning 1961–2022 from the CN05.1 gridded dataset (0.25° × 0.25° horizontal resolution), derived from over 2000 quality-controlled meteorological stations in China (Wu and Gao 2013). Daily mean sea-level pressure (SLP) and geopotential height fields for the same period are taken from the NCEP/NCAR Reanalysis 1 at 2.5° × 2.5° resolution.

The coupled model intercomparison project Phase 6 (CMIP6) provides the most comprehensive multi-model ensemble to date for climate change detection and attribution studies (Eyring *et al* 2016). This study makes use of large-ensemble historical simulations from the detection and attribution model intercomparison project (DAMIP) tier-1 experiments

(Gillett *et al* 2016). To ensure a sufficient sample size, we used six CMIP6 models participating in DAMIP, giving a total of 41 simulations. These six models were selected because each provides a least three ensembles with different external forcing experiments. Simulations for the period 1961–2014 include four forcing scenarios: historical all forcings (ALL, including both anthropogenic and natural forcings), anthropogenic GHG-only forcings, anthropogenic aerosol-only forcings (AER), and natural-only forcings (NAT, comprising solar variability and volcanic eruptions) (model details and ensemble sizes are listed in table S1). Monthly SAT and Pre from all model runs were bilinearly interpolated to the 0.25° × 0.25° grid of the CN05.1 observational dataset to facilitate direct comparison.

2.2. Method

2.2.1. Definition of CWCE

To quantify the severity of CWCE, this study defines a simple CWCE index (CWCEI) as follows:

$$\mathrm{CWCEI} = \frac{1}{2}\left(\frac{T'}{\sigma_T} - \frac{P'}{\sigma_P}\right) \qquad (1)$$

where and P' represent the anomalies of the area-weighted regional-mean February SAT and Pre time series over southern China (106.75–121.75 °E, 21.75–28.75 °N), and σ_T and σ_P are the corresponding standard deviation over the study period. All anomalies in this study refer to deviations relative to the 1991–2020 climatology mean. By standardising both variables before combining them, the index removes unit differences and gives equal weight to SAT and Pre anomalies. Smaller (more negative) CWCEI indicate more severe CWCE conditions.

2.2.2. Circulation analogue method

Identifying atmospheric circulation patterns similar to those observed in February 2022 is a key step in quantifying the contribution of atmospheric circulation to CWCEs. Here we adopt the constructed circulation analogue method originally developed by Deser *et al* (2016) to separate dynamic and thermodynamic contributions to winter SAT trends over North America. The approach was latter refined by Qian *et al* (2023) specifically for extreme-event attribution, making it well-suited for isolating the role of atmospheric circulation in driving individual high-impact events.

The procedure is as follows:

Step1: Nonlinear trends in daily SLP, SAT, and Pre for 1961–2021 are removed at each grid cell using a quadratic polynomial fit to minimise the influence of external forcing on the circulation analogue.

Step2: For each day in February 2022, detrended SLP anomalies (domain: 0°–60 °N, 80°–160 °E) are compared with all days falling within a ±30-d window (61 d total) centred on the corresponding calendar date during 1961–2021 (61 years). The Na days with the highest spatial correlation are retained as candidate analogues. Na was set to 100, meaning that for each day, the top 100 historical SLP anomaly patterns with the highest correlation coefficient were selected from the 61 × 61 historical patterns.

Step3: From these Na candidates, selecting Ns analogues are randomly drawn with replacement. Optimal weighting coefficients are derived by singular value decomposition that minimise the root-mean-square error between observed and analogue-composited SLP. The same weights are then applied to the corresponding SAT and Pre anomalies of the Ns analogues to produce one reconstructed field for the target day. This random sampling and reconstruction are repeated 1000 times. The 5th and 95th percentiles of the reconstructed fields are subsequently calculated, yielding a 95% confidence interval (CI).

The circulation contribution is quantified by comparing the distribution of the reconstructed CWCE index with two control experiments (Control-1, Control-M) that represent conditions expected under typical February circulation. Control-1 consists of 1000 ensembles in which, for each day in February 2022, one day is randomly selected from the same ±30-d calendar window during 1961–2021. Control-M uses the same random selection but excludes cases where any two selected days are separated by 2 d or fewer, thereby accounting for typical SLP persistence.

The circulation contribution (%) is calculated as:

$$\frac{\mathrm{median}\left(\mathrm{CWCEI}_{\mathrm{analogue}}\right) - \mathrm{median}\left(\mathrm{CWCEI}_{\mathrm{control-M}}\right)}{\mathrm{observed\ detrended\ CWCEI}_{2022}}. \qquad (2)$$

To estimate the thermodynamic influence of climate change, the circulation analogue reconstruction method is applied for 1961–1984 (with weaker climate change influence) and 1985–2021 (with stronger climate change influence). Notably, the reconstructed SAT and Pre fields were not detrended, which allows the thermodynamic impacts of climate change to be retained. The difference in reconstructed CWCE intensity between these two periods, under nearly identical circulation patterns, primarily reflects the thermodynamic effect associated with climate change (Stott *et al* 2016, Ye and Qian 2021).

2.2.3. Fraction of attribution risk

This study uses the fraction of attributable risk (FAR) and probability ratio (PR) to quantify the contribution of human activities to this CWCE (Fischer and Knutti 2015), defined as:

$$\mathrm{FAR} = 1 - \frac{P_0}{P_1} \tag{3}$$

$$\mathrm{PR} = \frac{P_1}{P_0} \tag{4}$$

where, P_0 and P_1 are the probabilities of occurrence of the CWCE in the counterfactual world (without or weaker human influence), and in the factual world (with or stronger human influence), respectively. The 95% CIs for FAR and PR in the analogue-based reconstruction were derived from the bootstrapping procedure in Step 3 of section 2.2.2. For the DAMIP simulations, they were derived by resampling with replacement 1000 times from the 54 year (1961–2014) time series of 41 ensemble members under external forcing, with the 5th and 95th percentiles of the resulting 10 000 estimates used as the confidence bounds.

3. Results

3.1. Analysis of southern China 2022 CWCEs

In February 2022, most regions in southern China experienced negative temperature anomalies (figure 1(a), with a regional average SAT of 6.18 °C and an anomaly of −3.42 °C (figure 1(c)). This ranks as the second lowest value since 1990, surpassed only by 2008. After detrending, the anomaly reach −4.1 °C, marking the lowest since 1961. Most regions exhibited positive precipitation anomalies, particularly in the southern areas (figure 1(b)). The regional average Pre was 5.23 mm d^{-1}, with an anomaly of 2.6 mm d^{-1}, marking the highest value since 1990. The detrended anomaly was also 2.6 mm d^{-1}, the third highest on record since 1961 (exceeded only by 1983 and 1985). Consequently, the CWCEI anomaly reached—2.4, making the second strongest event on record, surpassed only by 1983. The detrended anomaly reach −2.4, marking the most extreme event in the 1961–2022 record (figure 1(c)).

The 500 hPa geopotential height field in February 2022 exhibited pronounced positive anomalies over north of ∼50 °N across the Eurasian continent, with negative anomalies over most of China (figure 2(a)). At the boundary between these anomalies, anomalous easterly flow prevailed (blue dashed line in figure 2(b)), along with a marked weakening of the mid-latitude westerly jet. In contrast, near 30 °N, anomalous westerly winds appeared near 30° N (red solid line in figure 2(b)), associated with a locally strengthened jet core. This meridional dipole pattern favoured repeated southward intrusions of cold air from high latitudes, providing the primary dynamic driver for the extreme low temperatures observed over southern China.

The anomalous Eurasian geopotential height pattern was closely linked to a strongly positive phase of the North Atlantic oscillation and pronounced warm SST anomalies across the North Atlantic. This structure triggered a circumglobally Rossby wave train from the Barents Sea Siberia to China (figure 2(c)), creating a '−+−' height anomaly pattern that sustained and enhanced the Ural blocking high while deepening the East Asian trough, thus supporting recurrent cold air outbreaks into southern China.

Meanwhile, negative 500 hPa height anomalies over the northern Indian Ocean and Arabian Sea signalled an active southern branch trough, which enhanced moisture transport from the tropical Indian Ocean into southern China (figure 2(d)). In the northwestern Pacific, a prominent anomalous anticyclone produced southwesterly flow along its western side, carrying warm, moist air into the region. Near the surface (figure 2(e)), a broad anomalous high-pressure system in the north produced low-level northerly winds over southern China, intensifying cold advection near the ground.

In summary, the positive geopotential height anomalies at high latitudes brought cold air to southern China, while the strengthened trough at low latitudes continuously transported warm and moist flows to southern China, creating ideal conditions for persistent low temperatures coincident with abundant precipitation (often freezing), thereby sustaining the CWCE event throughout February 2022.

3.2. Impact of atmospheric circulation on CWCE

The performance of the constructed circulation-analogue method depends on three key parameters: the choice of circulation variable (here SLP), the spatial domain used for analogue selection, and the number of analogues drawn (Ns). To ensure robustness, we conducted systematic sensitivity experiments. Previous studies have used different variables for circulation-analogue selection; for example, Qian *et al* (2023) used SLP, while Jézéquel *et al* (2018) employed 500 hPa geopotential height. Considering the lack of reliable data for 500 hPa geopotential height over the Tibetan Plateau, this study selected SLP anomalies as the circulation variable throughout this study.

To test the sensitivity of the results to the choice of circulation domain, three different circulation regions were selected: a large region (10–70° N,

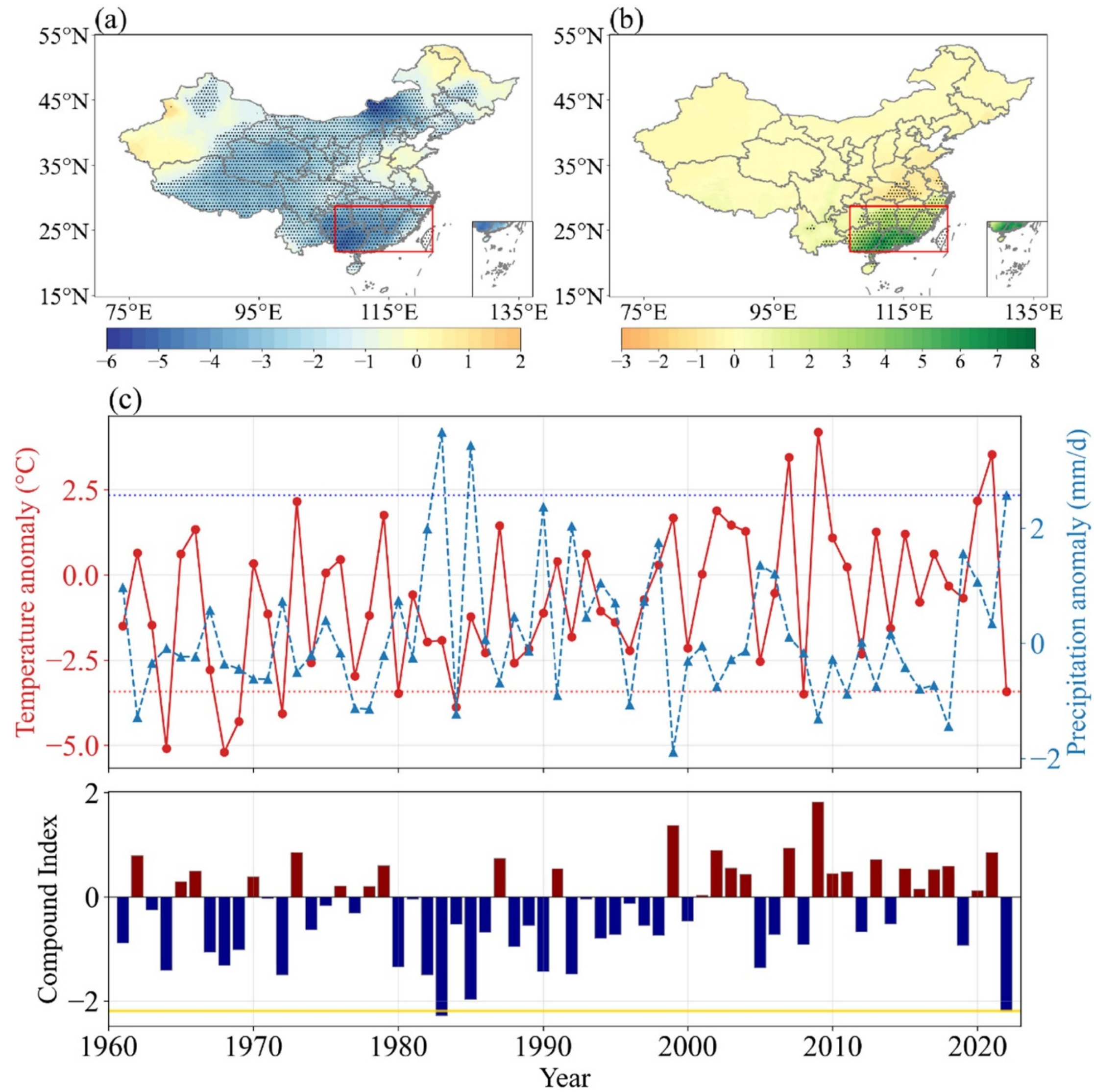

Figure 1. (a) SAT anomaly (°C) and (b) pre anomaly (mm/day) over China in February 2022, relative to the 1991–2020 climatology. Stippling denotes regions where anomalies exceed one standard deviation. The red rectangle marks the southern China. (c) Time series of area-weighted average February SAT (red), pre (blue), and CWCEI (bar) anomalies from 1961 to 2022. The horizontal lines represent the anomalies of February 2022 values.

90–130° E), a medium region (10–40° N, 100–120°E), and a small region (20–30° N, 105–120° E) (figure 3(a)). The analogue reconstruction was performed separately for each domain (figure 3(b)). The small domain produced a reconstructed CWCEI that was considerably more extreme than observed, with larger spreads in both temperature and precipitation anomalies and markedly wider uncertainty ranges. This indicates that an overly restricted domain overemphasises local noise and fails to capture the broader-scale circulation drivers. In contrast, the medium and large domains yielded reconstructed CWCEIs, SAT anomalies, and Pre anomalies that closely matched the observed February 2022 values. Given that the medium domain (10°–40° N, 100°–120° E) provided the best balance between fidelity to observations and reasonably narrow uncertainty, it was selected as the final circulation domain for all subsequent analyses.

To determine the optimal number of analogues sampled (Ns), reconstructions were performed for Ns ranging from 10 to 80 in Na = 100 candidate analogues (Qian *et al* 2023) (figure 3(c)). The constructed CWCEI most closely matched the February 2022 value when Ns = 60. Further examination shown that the reconstructed SAT anomalies was closet to observations at Ns = 60 (figure 3(d)), while the Pre anomaly matched best at Ns = 60, and 70 (figure 3(e)). Balancing these results, Ns = 60 was adopted for all subsequent analyses.

Following these parameter sensitivity tests, the constructed circulation-analogue method was

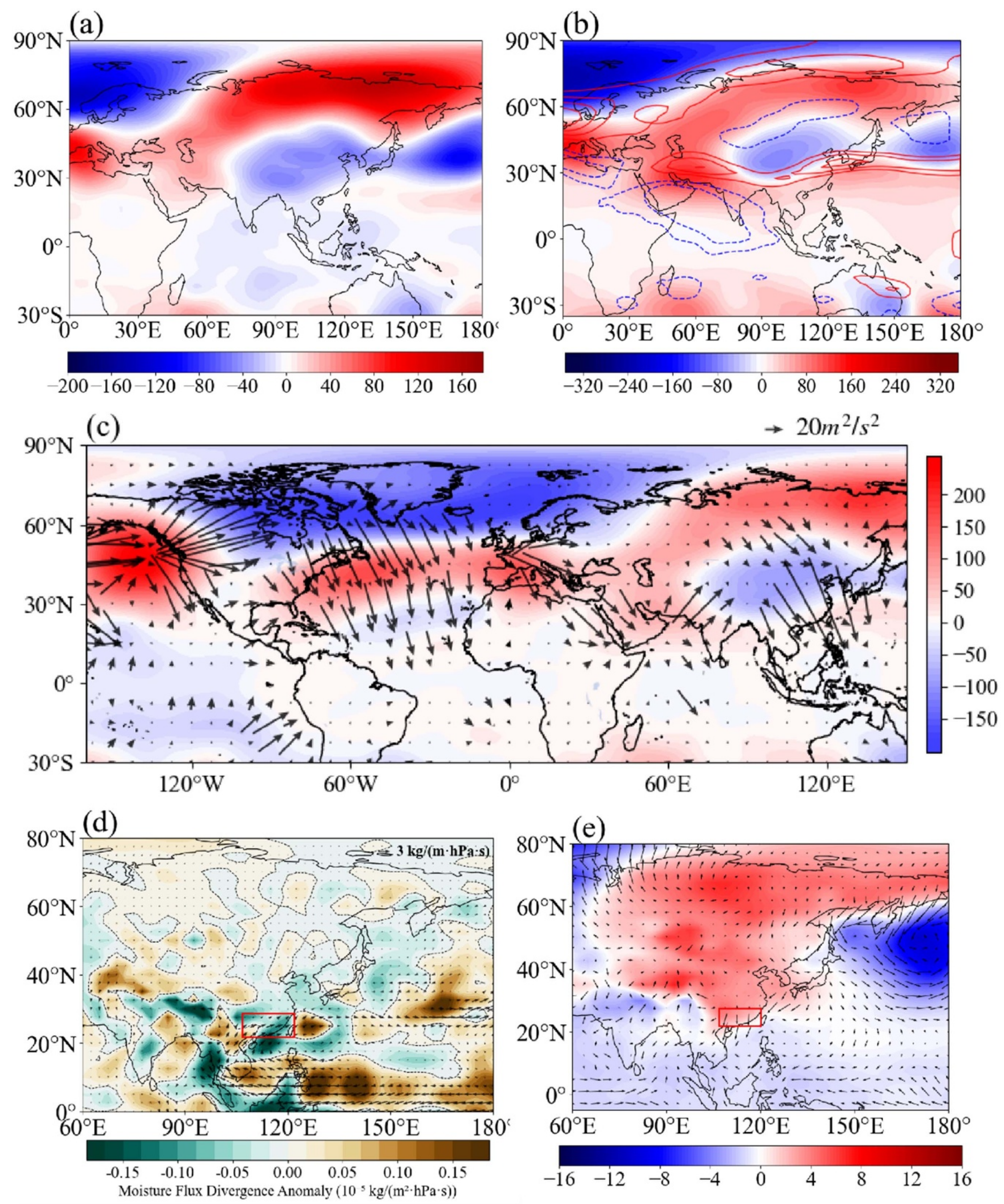

Figure 2. Atmospheric anomalies in February 2022 relative to the 1991–2020 climatology. (a) Anomalies 500 hPa geopotential height (shading; gpm). (b) Anomalies 200 hPa geopotential height (shading; gpm) and zonal wind (contour; m s^{-1}): red solid line and blue dashed line shows the positive and negative anomalies, respectively. (c) Anomalies 300 hPa geopotential height (shading; gpm) and wave activity flux (vector; $m^2 \cdot s^{-2}$). (d) Anomalies 700 hPa water vapour flux divergence (shaded; 10^{-5} kg $m^{-2} \cdot$ hPa · s) along with moisture flux (vector; kg m^{-1} · hPa · s). (e) Anomalies SLP anomaly (shading, hPa) and 10 m wind anomaly (vector; m · s^{-1}).

applied using a circulation domain of 10–40° N, 100–120° E, SLP anomalies as the circulation variable, and Ns = 60 randomly sampled analogues.

With these settings, the method successfully reproduces the main observed features of the February 2022 SLP, SAT and Pre anomaly fields (figure 4). Minor differences remain: the constructed negative SLP anomalies over Indian ocean are somewhat weaker and the positive anomalies over Eurasian continent are also weaker (figures 4(a) and (b)). Consequently, reconstructed Pre and SAT anomalies over southern China is weaker than observed (figures 4(c)–(f)). After comparing the detrended observed large-scale circulation anomalies (figure S1) with the reconstructed anomalies (figure S2), it is found that the reconstructed circulation does not fully reproduce all the features of the observed circulation anomalies. This is primarily because the circulation anomalies responsible for CWCEs in southern China are highly complex. Nevertheless, the

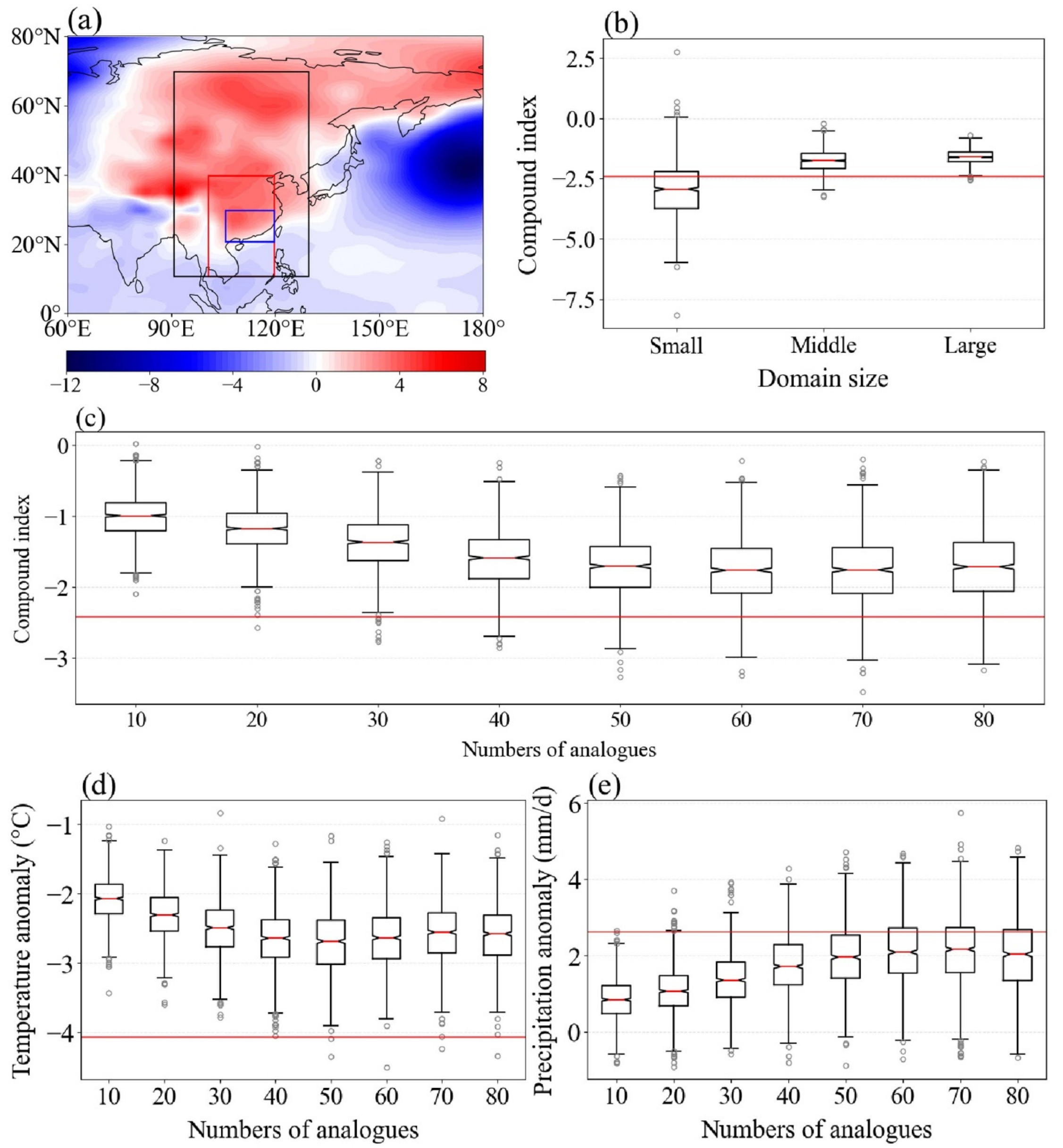

Figure 3. Sensitivity experiments for the circulation-analogue method. (a) Sea-level pressure detrended anomalies in February 2022 (shading; hPa), with the three tested circulation domains marked: small (blue rectangle), medium (red rectangle), and large (black rectangle). (b) Box-and-whisker plots of the reconstructed CWCEI obtained using the small, medium, and large domains with Na = 100 and Ns = 60. (c)–(e) Sensitivity of the reconstructed February 2022 (c) CWCEI, (d) SAT (°C), and (e) Pre (mm d^{-1}) to the Ns. Boxes and whiskers show the median, interquartile range, and 5th–95th percentiles across 1000 reconstructions. Note that raw SLP, SAT, and Pre were detrended prior to the reconstruction. The horizontal red lines in all panels indicate the observed detrended anomalies for February 2022.

reconstructed circulation successfully captures the main anomalous features, including: Positive geopotential height anomalies over eastern Siberia and negative anomalies over most of China (Peng *et al* 2021, Fu and Ding 2023), which provide favourable conditions for the southward transport of cold air. Negative geopotential height anomalies at 500 hPa over the Indian Ocean, together with positive anomalies in water vapour transport from the South China Sea at 700 hPa, provided favourable moisture conditions for this event.

The circulation-analogue reconstruction yields detrend anomalies for February CWCEI of −1.8, SAT of −2.6 °C, and Pre of 2.1 mm d^{-1} (figure 5(a)), compared with the observed value of −2.4, −4.1 °C, and 2.6 mm d^{-1}, respectively. In contrast, both control experiments (Control-1 and Control-M) produce near zero CWCEI, SAT, and Pre. The atmospheric circulation explains 73% (95% CI: 71%–76%) of the observed 2022 CWCE intensity. It accounts for 65% (95% CI: 64%–67%) of the extreme cold anomaly and 83% (95% CI: 80%–87%) of the

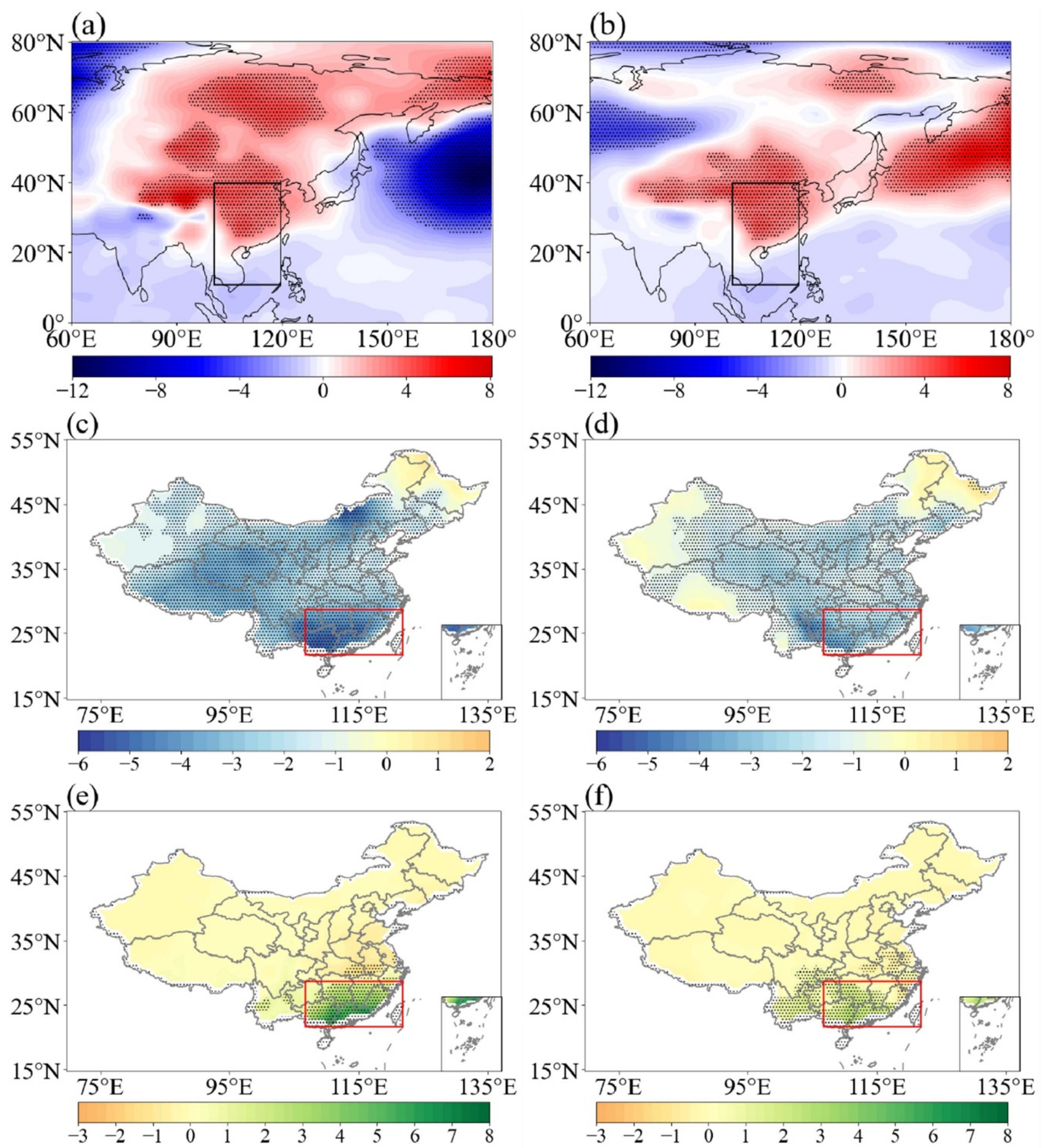

Figure 4. Comparison between observed and circulation-analogue reconstructed detrended anomalies fields for February 2022. The pattern of (a), (b) sea level pressure (hPa), (c), (d) SAT (°C), and (e), (f) Pre (mm d^{-1}) detrended anomalies for observation (a), (c), (e) and reconstruction (b), (d), (f) of February 2022, with stippled areas indicating anomalies exceeding one standard deviation. The black rectangle in (a), (b) denotes the circulation region; the red rectangle in (c)–(f) denotes the southern China.

extreme precipitation anomaly. These results demonstrate that the anomalous large-scale circulation in February 2022 was the dominant driver of both the persistent low temperature and heavy precipitation, and hence of the record-breaking CWCE over southern China.

3.3. Influence of climate change on the 2022-like CWCE

To isolate the thermodynamic effects of anthropogenic warming, we performed the circulation-analogue reconstruction separately for an earlier period characterised by weaker anthropogenic forcing (1961–1984) and a recent period with stronger forcing (1985–2021). In contrast to our prior analyses, the original trends were preserved in each periods prior to the reconstructions, the large-scale circulation was held fixed to match that observed anomalies in February 2022.

The results (figure 6) show that, under the same circulation pattern, the reconstructed CWCEI in the latter period is substantially less negative than in the previous period (figure 6(a)). This attenuation is driven almost entirely by higher reconstructed SAT in the latter period (figure 6(b)), whereas reconstructed Pre anomalies show little difference between the two periods (figure 6(c)). Thus, the observational evidence indicates that thermodynamic effect of climate

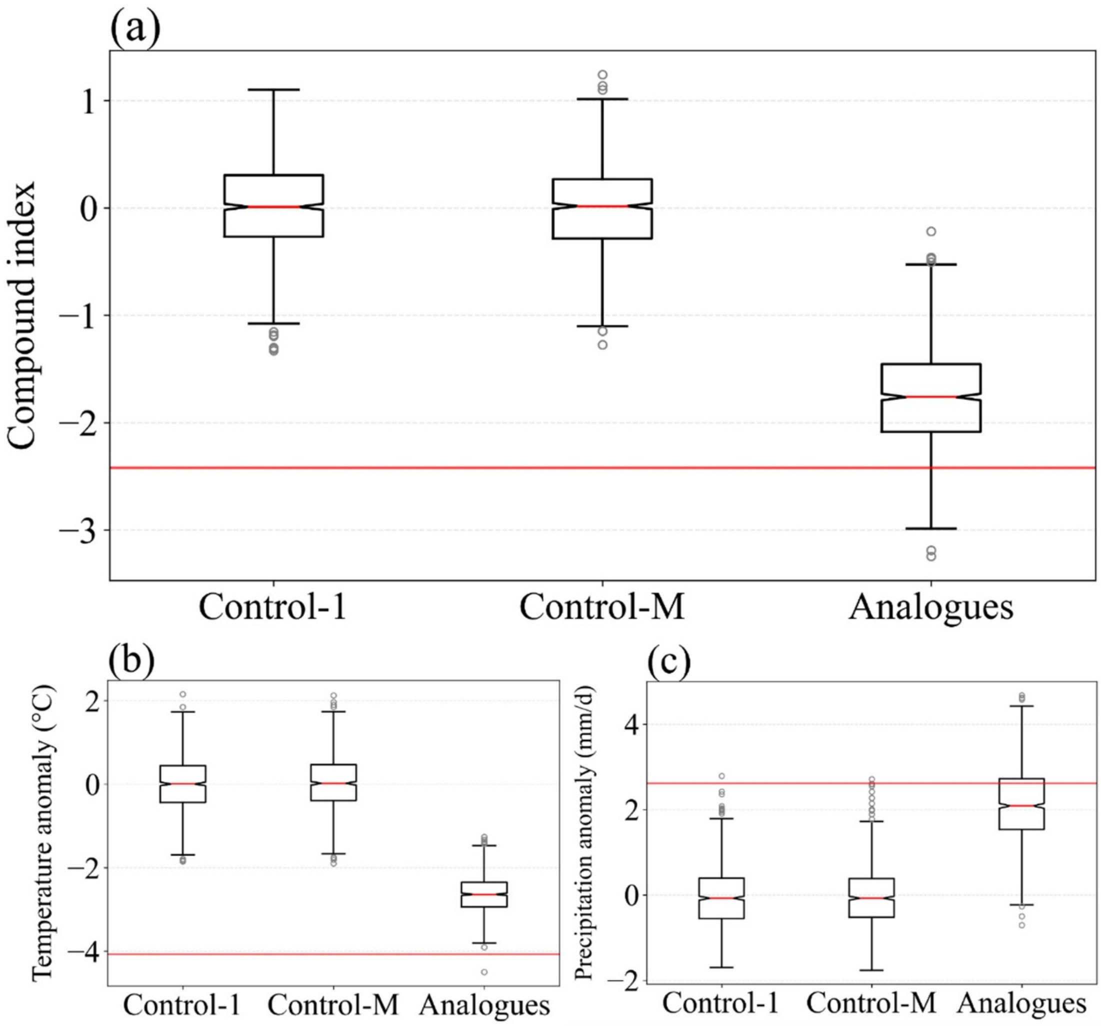

Figure 5. Contribution of atmospheric circulation to the event similar to February 2022 in southern China, derived from the constructed circulation-analogue method. The box-and-whisker plots of (a) CWCEI, (b) SAT anomaly (°C), and (c) pre anomaly (mm d^{-1}) for southern China, shown for the fully random control (Control-1), the persistent-adjustment random control (Control-M), and the circulation-analogue reconstruction (Analogues). Note that the raw SLP, SAT, and pre were detrended prior to the reconstruction. Horizontal red lines indicate the observed detrended anomalies of February 2022 values.

change primarily reduces the intensity of CWCE event by increasing SAT.

By fitting normal distributions to the 1000 reconstructed values of the CWCEI, SAT anomalies, and pre anomalies derived separately from the 1961–1984 and 1985–2021 (identical circulation conditions), we estimated the change in probability of events with intensities similar to or greater than the observed February 2022 event. In the earlier period (1961–1984), the probabilities of cold event, heavy precipitation event, and CWCEI event similar in intensity to those in February 2022 were 0.98, 0.30, and 0.66, respectively. The probability of cold event in the recent period (1985–2021) decreased to 0.62, the probability of heavy precipitation increased to 0.40, and the probability of CWCE decreased to 0.45. These findings indicate that the decrease in CWCEI intensity in the later period is primarily due to the reduction in the intensity of cold event. Further, PR and FAR are used to quantify the contribution of human activities. The PR is 0.68 (95% CIs: 0.64–0.73), meaning that the occurrence probability of such an event in recent period has decreased by approximately 32% (95% CIs: 27% to 36%) relative to earlier period. The corresponding FAR is −47% (95% CI: −57% to −37%), suggesting that the probability of the event would have been 47% (95% CIs: 37% to 57%) higher without climate change. This suppression is driven primarily by a PR of 0.63 (95% CI: 0.60–0.65) and a corresponding FAR of −60% (95% CI: −67% to −54%) for the cold event, reflecting strong thermodynamic warming that has substantially reduce the likelihood of cold events. This effect is only partly offset by a PR of 1.32 (95% CI: 1.19–1.45) and FAR of +24% (95% CI: +16% to +31%) for the extreme precipitation event, which arises from the increased atmospheric moisture capacity in a warmer climate that thus favours the occurrence of heavy precipitation.

We further employed CMIP6 models under different external forcings to investigate the influence

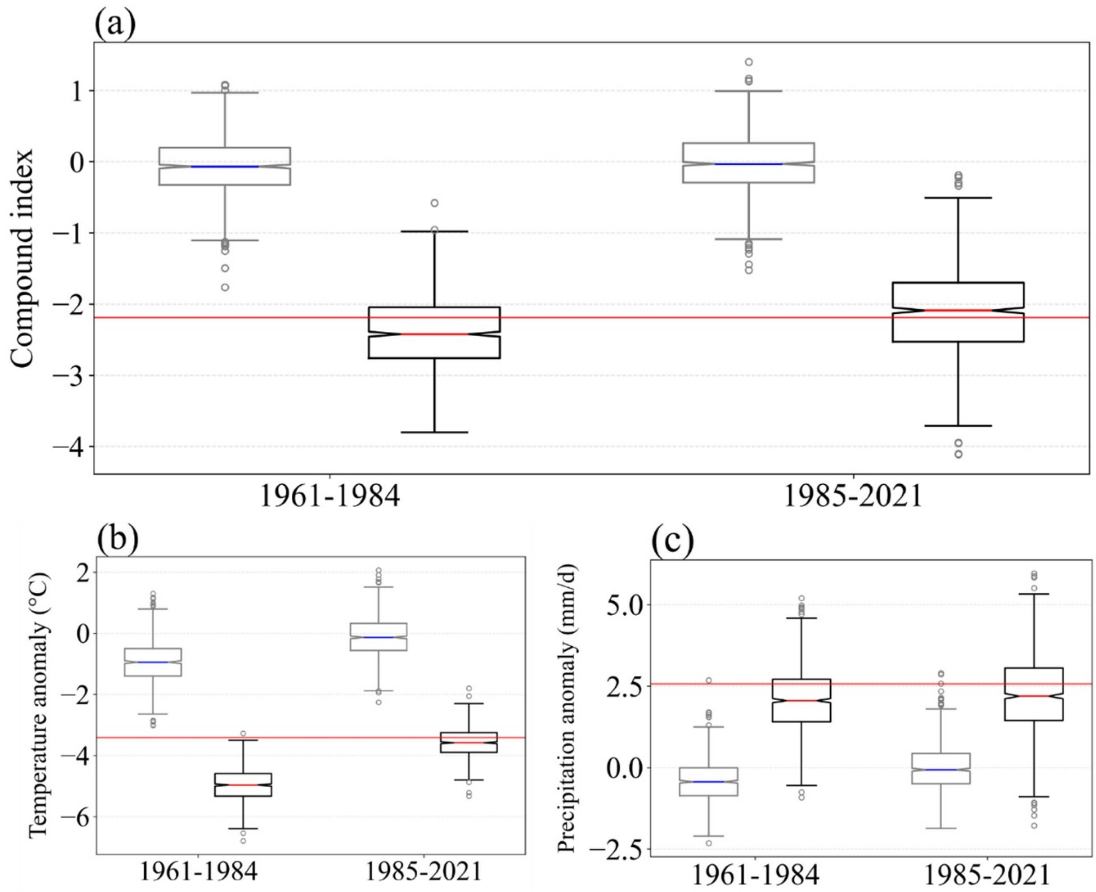

Figure 6. The intensity of event similar to the February 2022 in southern China during two periods (1961–1984 and 1985–2021) based on the circulation analogue method. The box-and-whisker plots of (a) CWCEI, (b) SAT anomaly (°C), and (c) pre anomaly (mm d^{-1}) calculated from 1000 reconstructions for southern China during 1961–1984 and 1985–2021. For each period, the light and dark box represents the random control group (Control-M) and the reconstructed analogues, respectively. Note that the raw SLP, SAT, and pre here were not been detrended prior to the reconstruction. Horizontal red lines indicate the observed anomalies of 2022 values.

of anthropogenic activities on the probability of a record-breaking CWCE event like the one observed in February 2022. Before conducting the formal attribution, we evaluated the performance of models in simulating the CWCE over Southern China. The multi-model ensemble mean under ALL forcing successfully reproduces the observed CWCWI since the 1961 (figure 7(a)), as the observational temporal series falls within the temporal range of the model simulations. Because variability is partially cancelled when averaging across ensemble members, the interannual variability in the ensemble mean is weaker than in the observation. From a single run perspective, the variability of CWCEI simulated by the model is stronger than that in the observations. The Kolmogorov–Smirnov (K–S) test confirm that the probability distributions of February CWCEI, SAT and Pre anomalies in the multi-model ensembles of ALL forcings are statistically indistinguishable from the observations ($p = 0.68$, $p = 0.09$, and $p = 0.30$, respectively; figures 7(b)–(d). These results demonstrate that the selected models are sufficiently reliable for probabilistic attribution of the 2022 CWCE event.

To comprehensively investigate the role of different anthropogenic forcing, we calculated the probability of observed February 2022-like CWCE under All forcing, GHG forcing, AER forcing, and NAT forcing during 1961–2014 (table 1). The PR (the ratio of the probability under anthropogenic forcing to that under NAT forcing) was used to quantify the contribution of human-induced forcing to the event's occurrence probability. Under ALL forcing, the occurrence probability of CWCEs is significantly reduced, with a PR of 0.79 (95% CIs: 0.67–0.96). This reduction is primarily attributed to the contribution from GHG, with a PR of 0.72 (95% CIs: 0.60–0.87). Although AER also decrease the CWCE occurrence probability, their 95% CI includes 1 (95% CIs: 0.85–1.19). For cold events and heavy precipitation events, ALL forcing reduces the occurrence probabilities of both cold events and heavy precipitation, but the 95% CIs for both include 1. Although aerosols can suppress precipitation through direct and indirect effects, the 95% CIs for the PR includes 1, indicating low statistical confidence in this effect. GHG forcing decrease the occurrence probability of cold events

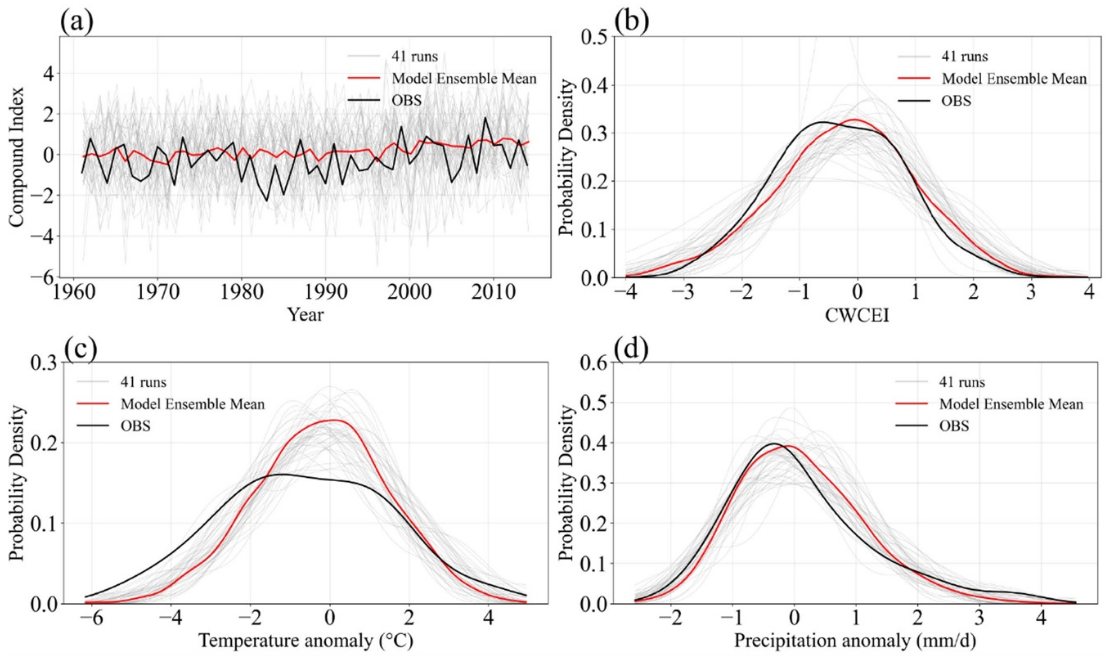

Figure 7. Evaluation of CMIP6 model performance for February SAT, pre and CWCE over southern China. (a) Time series of the February CWCEI over Southern China from 1961 to 2014, (b) probability density function of February CWCEI anomalies, (c) SAT anomalies (°C), and (d) pre anomalies (mm d^{-1}) for observation (black) and multi-model ensemble under ALL forcing (red) and individual ensemble members (grey lines).

Table 1. The risk ratio of CWCE, SAT, and pre event similar to February 2022 over southern China under ALL, GHG, and AER forcing, respectively.

	$PR_{ALL/NAT}$ (95% CI)	$PR_{AER/NAT}$ (95% CI)	$PR_{GHG/NAT}$ (95% CI)
CWCE	0.79 (0.67,0.96)	0.99 (0.85,1.19)	0.72 (0.60,0.87)
SAT	0.78 (0.52,1.15)	1.59 (1.17,2.13)	0.79 (0.54,1.08)
Pre	0.82 (0.52,1.18)	0.68 (0.37,1.03)	1.41 (0.94,2.06)

and increase that of heavy precipitation, but the 95% CIs also include 1. The results obtained from the 21 selected ensembles, which more accurately reproduce the observed CWCEI distribution (K–S test, $p > 0.5$), are similar to those from the full ensembles (table S2). This indicates the robustness of attribution results to the selection of the best-performing ensembles. At the regional scale, limited by the models' insufficient ability to simulate responses to different external forcings (Wang *et al* 2021), as well as the complex nonlinear interactions among different external forcings (Deng 2025, Zhao *et al* 2025), it is challenging to determine how anthropogenic influence CWCEs by altering cold events and heavy precipitation events. By comparing the probability of February 2022-like CWCE under ALL and NAT forcing, we quantified the contribution of anthropogenic influence. This yields a PR of 0.79 (95% CIs: 0.67–0.95), indicating that the anthropogenic forcing has reduced the probability of such CWCE by 21% (95% CIs: 5% to 33%). The corresponding FAR is −26% (95% CIs: −50% to −5%), meaning that the probability of this event would have been 26% (95% CIs: 5% to 50%) higher in the absence of anthropogenic influence.

4. Conclusion

This study investigates the record-breaking CWCE that affected southern China in February 2022. By applying the circulation-analogue approach to observational data, we quantified the contribution of large-scale atmospheric circulation and the thermodynamic effect of climate change to the event, and further used CMIP6 simulations under different external forcing to assess the anthropogenic influence. The Main conclusions are as follows:

(1) The February 2022 CWCE was the second most severe event in southern China since 1961, surpassed only by the 1983 event. The meridional dipole pattern in geopotential height, characterised by positive anomalies at high latitudes over Eurasia and negative anomalies across most

of China, promotes the southward advection of cold air. Concurrently, anomalously strong water vapour transport from the tropical Indian Ocean toward southern China enhances moisture availability in the region.

(2) By applying the circulation analogue reconstruction method to observation, atmospheric circulation contributed up to 73% (95% CI: 71%, 76%) of such CWCE. The thermodynamic effects of climate change reduced the probability of the CWCE by 32% (95% CI: 27%, 36%), mainly because climate change suppresses the occurrence of cold events.

(3) CMIP6 simulations confirms that anthropogenic influence decreased the probability of such CWCE by 21% (95% CI: 5%, 33%) from 1961 to 2014.

Based on observation, this study demonstrates that thermodynamic influence of climate change reduces the probability of CWCEs primarily by strongly decreasing the likelihood of cold extremes, while its influence on heavy precipitation is much less pronounced. These conclusions are consistent with previous studies (Qian *et al* 2018, Zhai *et al* 2018, Ye *et al* 2025). To isolate the thermodynamic contribution of climate change to CWCEs, we adopted a circulation-fixed approach that assumes negligible anthropogenic influence on large-scale circulation. This assumption is reasonable given the extreme difficulty of robustly quantifying anthropogenic effects on regional circulation anomalies at regional scales due to (1) the dominance of internal variability and correspondingly low signal-to-noise ratios (Stott *et al* 2010), (2) persistent systematic biases in current climate models regarding the position, intensity, and frequency of key circulation features (Sun *et al* 2024), and (3) non-linear interactions and partial cancellation between different external forcings (notably GHGs versus aerosols) substantially weaken the signal under ALL forcing (Deng *et al* 2020, Zhang *et al* 2025). It is worth noting that this method of fixing circulation patterns may not exclusively reflect the thermodynamic effects of climate change. Other factors, such as changes in underlying surface conditions (e.g. land use change), could also exert an influence, which warrants further investigation.

Despite its robustness, the circulation-analogue approach is inherently conditional and cannot fully account for potential anthropogenic modifications to circulation itself. The choice of circulation domain introduces some uncertainty. Due to the diverse circulation conditions contributing to CWCEs in southern China, reconstructing CWCEI solely based on small-area SLP circulation fields cannot fully reproduce the large-scale circulation anomaly features observed in February 2022, thereby introducing uncertainty into the attribution results. This necessitates the adoption of more accurate attribution methods, such as the storyline approach based on numerical simulations (Van der Wiel *et al* 2021, Wang *et al* 2023), to quantify the contribution of the thermodynamic effect of climate change under conditions that can fully simulate the circulation anomalies leading to February 2022.

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Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI. The daily reanalysis datasets derived from NCEP/NCAR Reanalysis 1 is available at https://psl.noaa.gov/data/gridded/data.ncep.reanalysis.html. The model simulations from CMIP6 can be downloaded from the Earth System Grid Federation (https://esgf-node.llnl.gov/projects/cmip6/).

The data that support the findings of this study are openly available at the following URL/DOI: https://doi.org/10.5281/zenodo.18884257.

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